Capstone Coding Project Background

Securing the Mobile-First Frontier: Assessing Socio-Technical Vulnerabilities and Fraud in Kenya's Mobile Money Ecosystem

E-Portfolio Link: https://github.com/Ojnymbok/Ojnyambok_Portfolio

Background

The rapid expansion of mobile telecommunications across Africa has driven financial inclusion to remarkable levels. In sub-Saharan Africa, mobile money has become the dominant way millions of people transact. Services like Safaricom's M-Pesa in Kenya have led the way, with millions relying on these platforms daily (GSMA, 2024/2025). However, this explosive growth has often outpaced regulation and security measures. Because these services are tied directly to a phone number (MSISDN) rather than a traditional bank account, they create unique vulnerabilities that cybercriminals are quick to exploit (Mudiri, 2013).

The Problem

Mobile money fraud is a critical socio-economic issue in Kenya. With penetration rates exceeding 77%, these platforms are lifelines for many families, yet they remain prime targets. Reports suggest that a significant portion of users have lost money to scams, eroding trust in the system (Central Bank of Kenya, 2023).

A typical attack pattern involves:

```
[Social Engineering / Vishing —> SMS / 2FA Interception —>
Unauthorized SIM Swap —> Account Takeover & Fund Drain]
```

Fraudsters use vishing or fake SMS messages to trick users into revealing PINs. They then collude with MNO agents to perform an unauthorized SIM swap, taking control of the victim's line and bypassing security. Key structural challenges include:

- **Ecosystem Interdependency:** Shared APIs between banks, telcos, and fintechs mean a vulnerability in one link can compromise the entire chain (Musyoka & Mose, 2024).
- **Skills and Awareness Gap:** Users often lack basic security awareness, and local cybersecurity expertise remains scarce (Enact Africa, 2021).
- **Legacy Infrastructure:** Heavy reliance on USSD and feature phones limits modern cryptographic protections.

Feature Phone Vulnerabilities and Shared Device Realities

A major security hurdle is the widespread use of low-cost feature phones (locally referred to as *Kabambe*). Unlike smartphones with app-based security and biometrics, many Kenyans interact with mobile money only through USSD menus and SMS. These channels lack end-to-end encryption, making them vulnerable to interception and ThinSIM overlay attacks.

Furthermore, device sharing in low-income or rural households is common. Multiple users sharing a single physical device breaks standard security assumptions around device fingerprinting and behavioural analysis (Enact Africa, 2021).

MNO Mitigations and Fraud Prevention

Mobile Network Operators (MNOs) have responded with practical, layered defences. For SIM swaps, they require strict in-person identity verification cross-checked against government ID databases (GSMA, 2024/2025; YouVerify 2026).

```
[SIM Swap Request Initiated] —> [API Alerts Financial Partners] —>
[48-72 Hour Transaction Freeze] —> [In-Person ID Verification
Required]
```

MNOs also share real-time alerts with financial partners using APIs like Safaricom's SIM-Swap-Check to flag high-risk transactions (Potentash, 2023). Additionally, user-controlled tools like USSD code *100*100# allow subscribers to self-lock their lines against unauthorized swaps.

It is for the above reasons that my project is developing a cyber security mechanism for Sawacom Ltd, fictitious MNO in Kenya. In developing the complementary Sawacom Ltd cybersecurity simulation, several of these controls—identity verification, cool-down logic, velocity checks, and USSD locking—were modelled to demonstrate their practical efficacy. Addressing mobile money fraud requires combining robust technical solutions with a deep understanding of local socio-cultural realities.

Overall, this research reinforced for me the value of combining technical solutions with an understanding of local socio-cultural realities. Addressing mobile money fraud in Kenya requires not only better code and APIs but also continued investment in awareness, skills development, and user-centric design.

Here is the link to the project: https://github.com/Ojnymbok/Ojnyambok_Portfolio

References

Central Bank of Kenya. (2023) *Mobile Money Statistics and Reports*.

Enact Africa. (2021) *The Impact of Communication Infrastructure and Digitisation on Organised Crime in Africa*.

GSMA. (2024/2025) *State of the Industry Report on Mobile Money*.

Mudiri, J. (2013) *Fraud in Mobile Financial Services*. MicroSave Consulting.

Musyoka, J. M. and Mose, T. (2024) 'Zero Trust Maturity Model and Cyber Resilience in Mobile Money Providers in Nairobi City County, Kenya', *International Journal of Social Science, Management and Entrepreneurship*, 8(3), pp. 1037-1051.

Potentash. (2023) *How Safaricom's SIM-Swap-Check Anti-Fraud Solution For Banks Will Help Protect You From Scammers*. Potentash Africa

YouVerify. (2026) *Mobile Money Fraud Prevention in Kenya: A Compliance Guide for Banks and Fintechs 2026*.